

3月28日 周六

微积分，共5题。

岁月流转，唯有思念与归风不变。

祝顺利！

1. $\lim_{x \rightarrow 0} \left(\frac{\arcsin x}{x} \right)^{\frac{1}{\sin^2 x}}$

$$A = \lim_{x \rightarrow 0} e^{\frac{1}{\sin^2 x} \ln \left(\frac{\arcsin x}{x} - 1 + 1 \right)}$$

$$\frac{1}{\sin^2 x} \ln \frac{\arcsin x}{x} - 1 = \lim_{x \rightarrow 0} \frac{x + \frac{x^3}{6} - x}{x^2} = \frac{1}{6}$$

$$\text{故 } A = e^{\frac{1}{6}}$$

$$\Delta \arcsin x = x + \frac{x^3}{6} + o(x^3)$$

2. $\lim_{x \rightarrow 0} \frac{\int_0^x [e^{(t-x)^2} - 1] \sin t \, dt}{x^2(e^{x^2} - 1)}$

$$A \xrightarrow[t=du]{t-x=u} \lim_{x \rightarrow 0} \frac{\int_{-x}^0 (e^{u^2} - 1) \sin(u+x) \, du}{x^4} = \lim_{x \rightarrow 0} \frac{\int_{-x}^0 (e^{u^2} - 1) \sin u \, du + \sin x \int_{-x}^0 (e^{u^2} - 1) \cos u \, du}{x^4}$$

$$\stackrel{\Delta}{=} \lim_{x \rightarrow 0} \frac{(e^{x^2} - 1) \sin(-x) + \cos x \int_{-x}^0 (e^{u^2} - 1) \cos u \, du + (e^{x^2} - 1) \sin x \cdot \cos x}{4x^3}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 - x}{4x^3} + \lim_{x \rightarrow 0} \frac{\int_{-x}^0 (e^{u^2} - 1) \cos u \, du}{4x^3} + \lim_{x \rightarrow 0} \frac{x^3}{4x^3}$$

$$\stackrel{\Delta}{=} \lim_{x \rightarrow 0} \frac{\cos x (e^{x^2} - 1)}{12x} = \frac{1}{12}$$

3. $\int_0^1 \ln \frac{1}{1-x^2} \, dx$

$$2 = -\int_0^1 (\ln(1+x) + \ln(1-x)) \, dx$$

$$= \left[(1+x) \ln(1+x) - (1+x) \right] \Big|_0^1 + \left[(1-x) \ln(1-x) - (1-x) \right] \Big|_0^1$$

$$= -1 - (2 \ln 2 - 2) + 1$$

$$= 2 - 2 \ln 2$$

Δ 本题题目打错了， $\int_0^1 \ln \frac{1}{1-x} \, dx$ ，可令 $t=1-x$ 解决。

4. 计算 $\int_1^e \cos(\ln x) \, dx$

$$\text{令 } t = \ln x, \quad x = e^t, \quad dx = e^t \, dt$$

$$2 = \int_0^1 \cos t \cdot e^t \, dt = \frac{1}{2} e^t (\sin t + \cos t) \Big|_0^1 = \frac{1}{2} (e \sin 1 + e \cos 1 - 1)$$

5. (思考题) 设函数 $f(x)$ 在 $(-1, 1)$ 上二阶可导, $f(0) = 1$, 且当 $x \geq 0$ 时, $f(x) \geq 0$, $f'(x) \leq 0$, $f''(x) \leq f(x)$, 证明: $f'(0) \geq -\sqrt{2}$

证: $f(x) = f(0) + f'(0)x + \frac{f''(\xi)}{2}x^2 \geq 0$. 且 $f'(x) \leq f(x)$, $f(x) \searrow$, $x \geq 0$

$$f'(0) \geq -\frac{f(0)}{x} - \frac{f''(\xi)}{2}x \geq -\frac{f(0)}{x} - \frac{f(x)}{2}x \geq -\frac{f(0)}{x} - \frac{f(0)}{2}x = -\frac{1}{x} - \frac{1}{2}x$$

故 对 $\forall x \in (0, 1)$, $-\frac{1}{x} - \frac{x}{2} \leq f'(0)$

$$\text{而 } \left(-\frac{1}{x} - \frac{x}{2}\right)_{\max} = -\sqrt{2}.$$

故 $f'(0) \geq \left(-\frac{1}{x} - \frac{x}{2}\right)_{\max} = -\sqrt{2}$. 证毕.